Topic 1: Acid / Base Disorders

EMERGENCY MEDICINE CLINICAL PRACTICE & BOARD SPECIFICATION

PATHOPHYSIOLOGY & ETIOLOGY

Primary metabolic or respiratory disturbances. Anion Gap = Na - (Cl + HCO3). Normal = 8-12 mEq/L. High AG due to MUDPILES: Methanol, Uremia, DKA, Paraldehyde, Iron/INH, Lactic acidosis, Ethylene glycol, Salicylates.

CLINICAL PRESENTATION & PHYSICAL EXAM

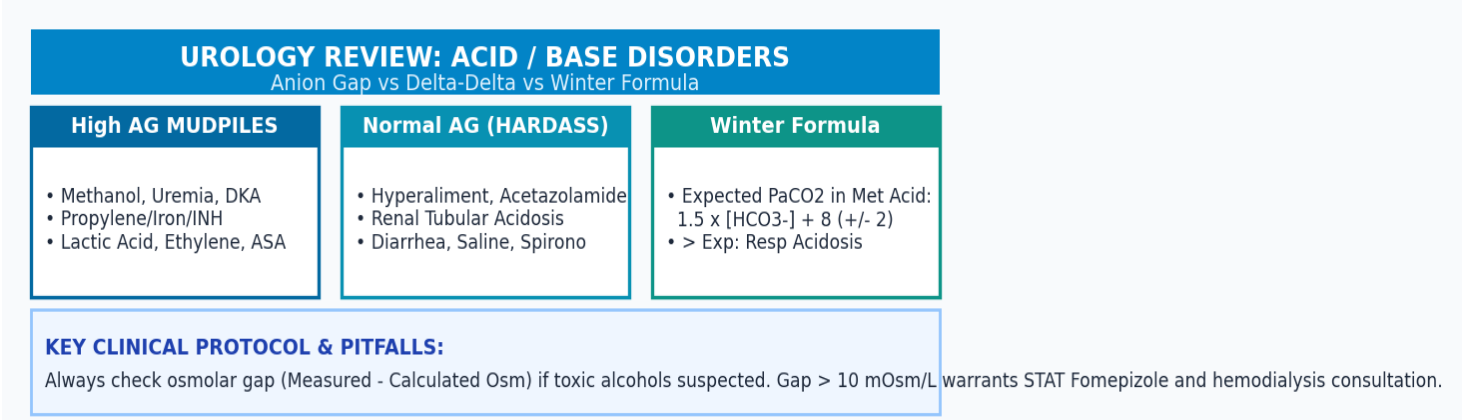
• **Presentation:** Tachypneic (Kussmaul breathing in DKA), altered mental status, vomiting, shock, or neuromuscular irritability in alkalosis.
• **Physical Exam:** Deep rapid breathing, signs of volume depletion, hypotension, altered sensorium, asterixis in severe uremic acidosis.

DIAGNOSTIC STUDIES & LABORATORY TRIAGE

STAT ABG/VBG, BMP with calculated Anion Gap, delta-delta ratio for mixed disorders, serum lactate, toxicology screen.

MANAGEMENT & EMERGENCY RESUSCITATION PROTOCOLS

Treat underlying cause (insulin for DKA, dialysis for severe uremia/toxic alcohols). Judicious sodium bicarbonate only for severe acidemia (pH < 7.1) or bicarb-responsive toxins.



HIGH-YIELD BOARD PEARL: In high anion gap acidosis, check the Delta-Delta ratio [ΔAG / ΔHCO3-]: <0.4 indicates mixed High AG and Normal AG acidosis; >2.0 indicates concurrent metabolic alkalosis.

Topic 2: Acute Renal Failure (AKI)

EMERGENCY MEDICINE CLINICAL PRACTICE & BOARD SPECIFICATION

PATHOPHYSIOLOGY & ETIOLOGY

Abrupt decrease in GFR. Categorized into Prerenal (hypoperfusion, FeNa < 1%, BUN:Cr > 20:1), Intrinsic (ATN from ischemia/toxins, AIN, muddy brown granular casts), and Postrenal (obstruction).

CLINICAL PRESENTATION & PHYSICAL EXAM

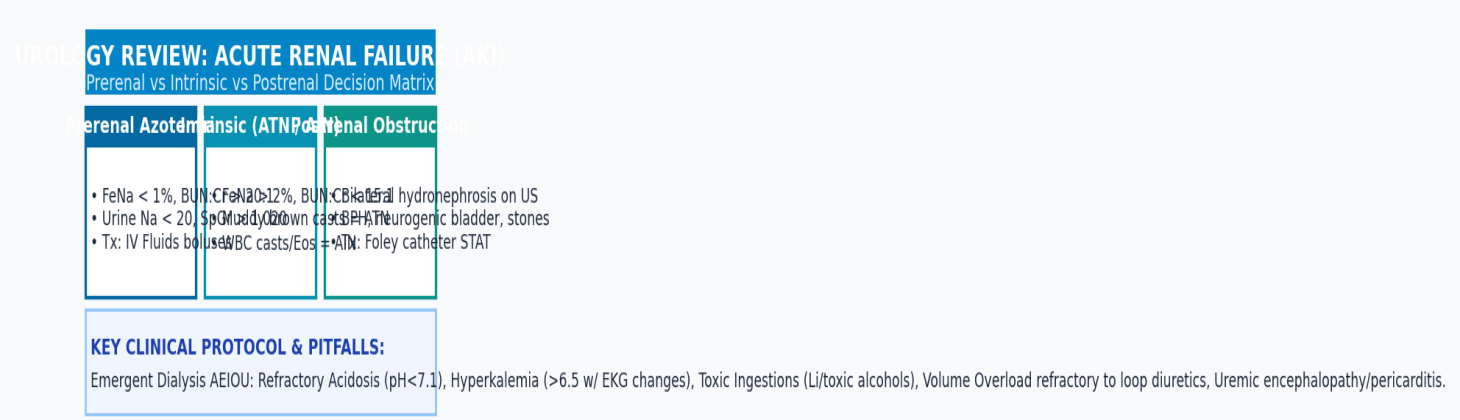
- **Presentation:** Oliguria/anuria or normal urine output, peripheral edema, nausea, anorexia, encephalopathy, dyspnea from volume overload.
- **Physical Exam:** Fluid overload (JVD, pulmonary crackles, pedal edema) or dehydration. Pericardial friction rub in uremic pericarditis.

DIAGNOSTIC STUDIES & LABORATORY TRIAGE

Serum creatinine, BUN, electrolytes (STAT K+ level), urinalysis with microscopy (muddy brown = ATN; WBC casts/eosinophils = AIN; RBC casts = Glomerulonephritis), Renal US.

MANAGEMENT & EMERGENCY RESUSCITATION PROTOCOLS

Prerenal: isotonic crystalloid resuscitation. Postrenal: Foley catheter or emergent nephrostomy. Emergent Hemodialysis indications: AEIOU (Acidosis, Electrolytes/K+, Ingestion, Overload, Uremia).



HIGH-YIELD BOARD PEARL: Muddy brown granular casts are pathognomonic for Acute Tubular Necrosis (ATN). FeNa < 1% indicates Prerenal azotemia; FeNa > 2% indicates ATN (unless patient is on loop diuretics, then use FeUrea < 35%).

Topic 3: Cystitis (Uncomplicated & Complicated)

EMERGENCY MEDICINE CLINICAL PRACTICE & BOARD SPECIFICATION

PATHOPHYSIOLOGY & ETIOLOGY

Infection of the bladder mucosa, predominantly ascending enteric pathogens. Most common organism: Escherichia coli (75-95%), followed by Klebsiella pneumoniae, Proteus mirabilis, and Staph saprophyticus.

CLINICAL PRESENTATION & PHYSICAL EXAM

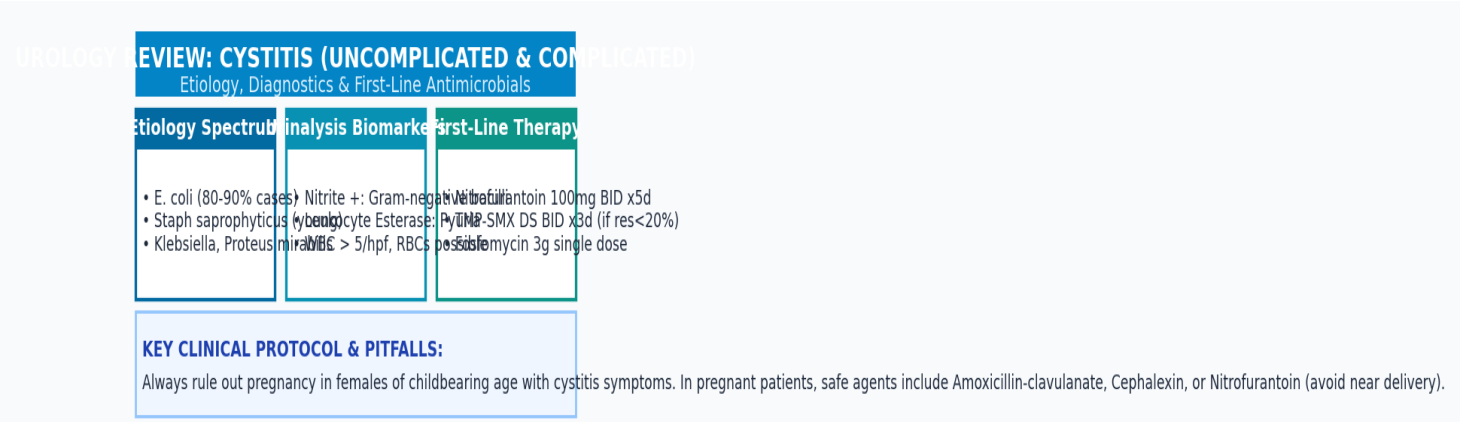
• **Presentation:** Dysuria, frequency, urgency, suprapubic discomfort, hematuria. Absence of fever, flank pain, or CVA tenderness (which indicates upper tract involvement).
• **Physical Exam:** Suprapubic tenderness without costovertebral angle (CVA) tenderness. Normal pelvic exam (excludes PID/vaginitis).

DIAGNOSTIC STUDIES & LABORATORY TRIAGE

Urinalysis: positive leukocyte esterase and positive nitrites (gram-negative rods). Urine microscopy: pyuria (>5 WBCs/HPF). Urine culture for complicated cases.

MANAGEMENT & EMERGENCY RESUSCITATION PROTOCOLS

Uncomplicated: Nitrofurantoin 100 mg BID x 5d OR TMP-SMX DS BID x 3d OR Fosfomycin 3g single dose. Complicated (males, pregnancy, immunosuppression): Fluoroquinolone or Cephalosporin x 7-14d.



HIGH-YIELD BOARD PEARL: Asymptomatic bacteriuria requires treatment in PREGNANCY and prior to urologic procedures to prevent pyelonephritis and preterm labor. Nitrofurantoin is contraindicated at term (weeks 36-42) due to hemolytic anemia risk in newborn.

Topic 4: Epididymitis & Orchiepididymitis

EMERGENCY MEDICINE CLINICAL PRACTICE & BOARD SPECIFICATION

PATHOPHYSIOLOGY & ETIOLOGY

Retrograde ascent of pathogens from urethra/bladder into epididymis. Age < 35: Chlamydia trachomatis & Neisseria gonorrhoeae. Age > 35: enteric coliforms (E. coli, Pseudomonas).

CLINICAL PRESENTATION & PHYSICAL EXAM

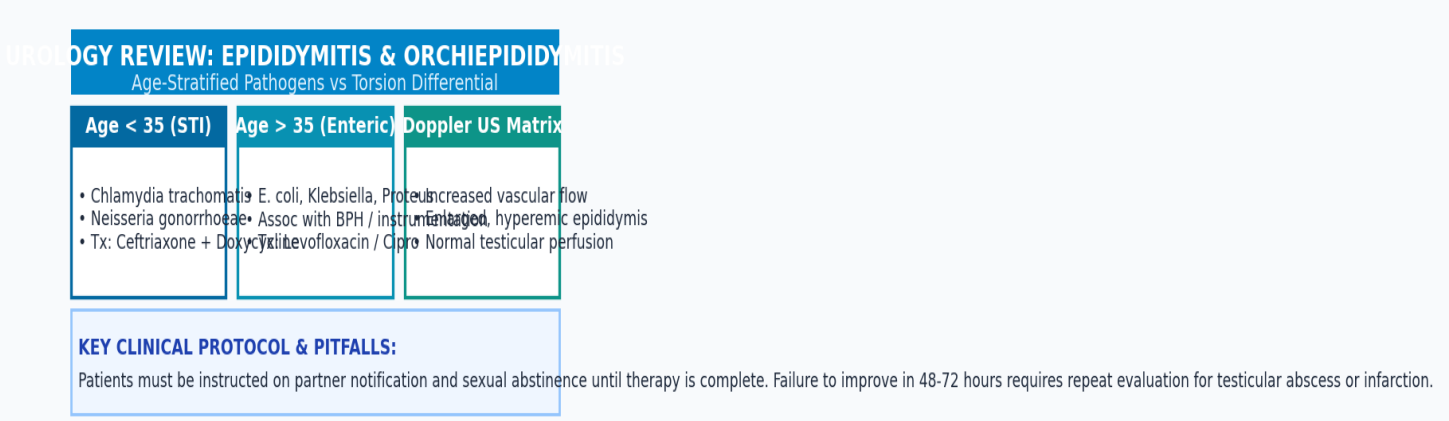
• **Presentation:** Gradual onset of unilateral scrotal pain, swelling, dysuria, urethral discharge, fever.
• **Physical Exam:** Swollen, tender epididymis distinct from testis early on. Positive Prehn sign (pain relief with scrotal elevation - unreliable). Normal intact cremasteric reflex (differentiates from torsion).

DIAGNOSTIC STUDIES & LABORATORY TRIAGE

Scrotal color Doppler US (shows increased blood flow / hypervascularity, excludes torsion). NAAT for GC/Chlamydia, urinalysis with culture.

MANAGEMENT & EMERGENCY RESUSCITATION PROTOCOLS

Age < 35: Ceftriaxone 500 mg IM single dose + Doxycycline 100 mg BID x 10d. Age > 35: Levofloxacin 500 mg daily x 10d OR Ciprofloxacin. Scrotal support, ice, NSAIDs.



HIGH-YIELD BOARD PEARL: Never rely on Prehn sign alone to rule out testicular torsion. An intact cremasteric reflex strongly favors epididymitis over torsion, but emergent Scrotal Doppler US is mandatory if any clinical doubt exists.

Topic 5: Fluid and Electrolyte Disorders (Potassium & Sodium)

EMERGENCY MEDICINE CLINICAL PRACTICE & BOARD SPECIFICATION

PATHOPHYSIOLOGY & ETIOLOGY

Hyperkalemia (K > 5.0) causes cardiac conduction arrest. Hypokalemia (K < 3.5) causes arrhythmias, U waves. Hyponatremia (Na < 135) categorized by volume status; rapid correction risks ODS. Hypernatremia causes neuronal shrinkage.

CLINICAL PRESENTATION & PHYSICAL EXAM

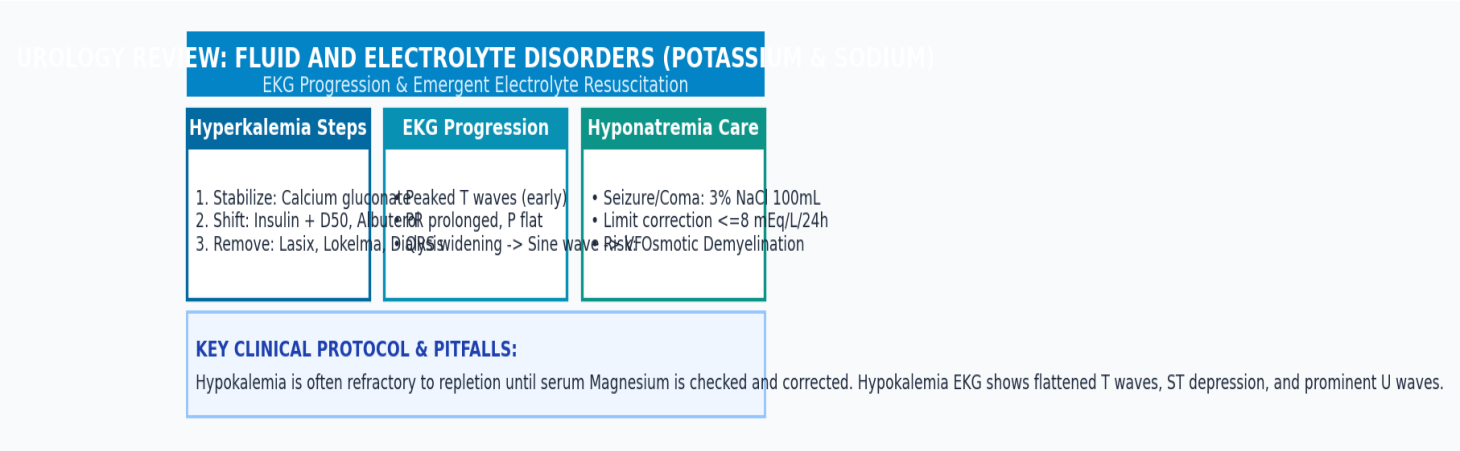
• **Presentation:** Hyperkalemia: weakness, flaccid paralysis, paresthesias. Hyponatremia: nausea, headache, seizures, coma (< 120). Hypokalemia: cramps, constipation, muscle necrosis.
• **Physical Exam:** Hyporeflexia, flaccid weakness, altered mentation, signs of volume overload or severe dehydration.

DIAGNOSTIC STUDIES & LABORATORY TRIAGE

STAT Serum Electrolytes, BUN/Cr, calculated Serum Osmolality, Urine Na and Osmolality, 12-lead EKG.

MANAGEMENT & EMERGENCY RESUSCITATION PROTOCOLS

Hyperkalemia w/ EKG changes: 1) Stabilize membrane: IV Calcium Gluconate (10 mL 10%); 2) Shift K: Regular Insulin 10U IV + D50W 50 mL + nebulized Albuterol; 3) Eliminate K: Furosemide, Patiromer, Hemodialysis. Severe symptomatic Hyponatremia (seizures): 3% Hypertonic Saline 100 mL bolus (max increase 6-8 mEq/L in 24h to prevent Osmotic Demyelination Syndrome).



HIGH-YIELD BOARD PEARL: Calcium stabilizes the cardiac membrane in hyperkalemia within 1-3 minutes but DOES NOT lower serum potassium. In severe hyponatremia, never exceed a sodium correction rate of 8 mEq/L per 24 hours to avoid Osmotic Demyelination Syndrome (central pontine myelinolysis).

Topic 6: Glomerulonephritis (Nephritic Syndrome)

EMERGENCY MEDICINE CLINICAL PRACTICE & BOARD SPECIFICATION

PATHOPHYSIOLOGY & ETIOLOGY

Immune-mediated inflammation of glomeruli resulting in capillary damage, hematuria, proteinuria, and azotemia. Causes: Post-streptococcal (PSGN), IgA nephropathy (Berger disease), Lupus nephritis, ANCA vasculitis, Goodpasture syndrome.

CLINICAL PRESENTATION & PHYSICAL EXAM

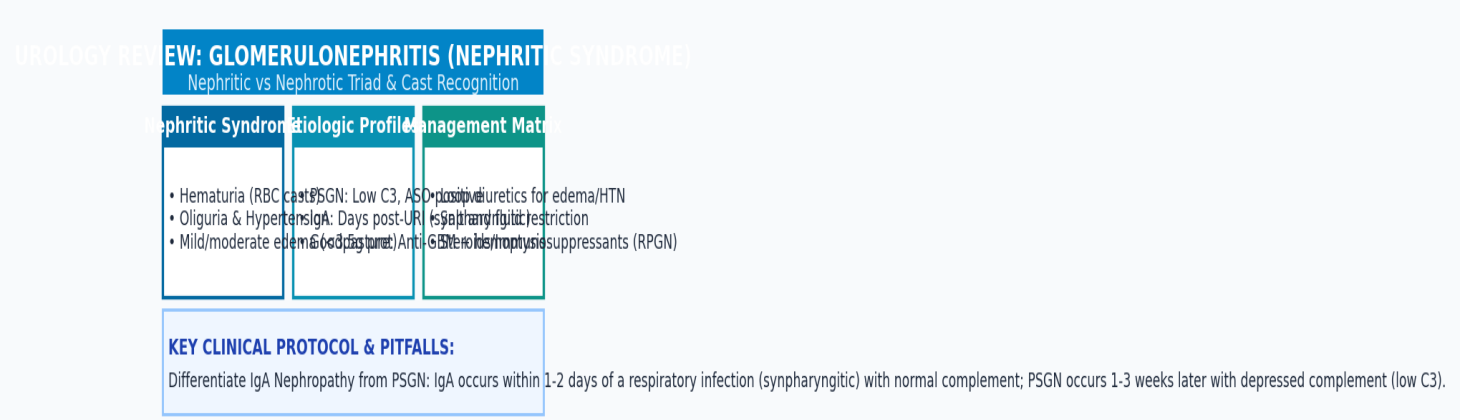
- **Presentation:** Dark, cola-colored (tea-colored) urine, facial and periorbital edema, oliguria, shortness of breath, acute hypertension.
- **Physical Exam:** Hypertension, periorbital edema (especially morning), peripheral pitting edema, pulmonary crackles from fluid retention.

DIAGNOSTIC STUDIES & LABORATORY TRIAGE

Urinalysis showing Dysmorphic RBCs and RBC casts (hallmark). Proteinuria (< 3.5 g/24h). Elevated BUN/Cr. Low C3/C4 (PSGN, Lupus). Elevated ASO titer or Anti-DNase B in PSGN.

MANAGEMENT & EMERGENCY RESUSCITATION PROTOCOLS

Blood pressure and volume control: Loop diuretics (Furosemide) for volume overload/HTN. ACE inhibitors / ARBs long-term. High-dose pulse Corticosteroids for rapidly progressive glomerulonephritis (RPGN). Emergent dialysis if uremic or refractory hyperkalemia.



HIGH-YIELD BOARD PEARL: RBC casts and dysmorphic RBCs on urine microscopy are definitive evidence of glomerular bleeding. PSGN typically occurs 1-3 weeks after streptococcal pharyngitis or 3-6 weeks after impetigo and features low C3 levels.

Topic 7: Hematuria (Gross & Microscopic Triage)

EMERGENCY MEDICINE CLINICAL PRACTICE & BOARD SPECIFICATION

PATHOPHYSIOLOGY & ETIOLOGY

Presence of RBCs in urine: Microscopic (>= 3 RBCs/HPF) or Gross. Sources: Glomerular (dysmorphic RBCs, RBC casts, brown/cola color, proteinuria) vs Non-glomerular (normal morphology, clots, pink/red color: stones, trauma, malignancy, infection).

CLINICAL PRESENTATION & PHYSICAL EXAM

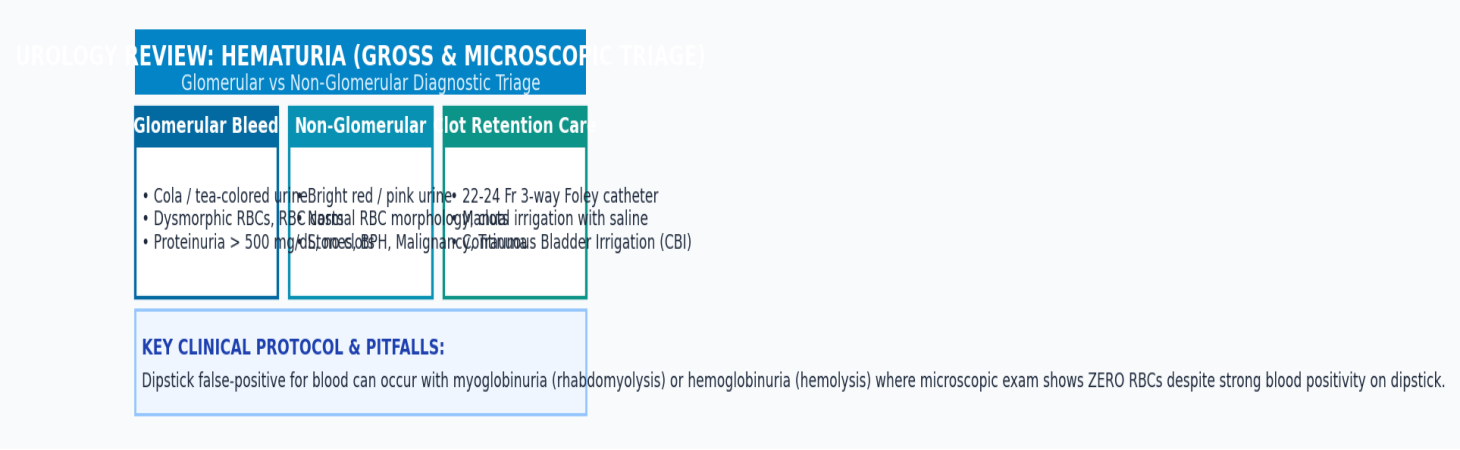
• **Presentation:** Painless gross hematuria is Bladder Cancer until proven otherwise. Painful hematuria suggests nephrolithiasis, UTI, or papillary necrosis.
• **Physical Exam:** Flank tenderness (nephrolithiasis, pyelonephritis), suprapubic tenderness (cystitis), enlarged prostate or pelvic mass.

DIAGNOSTIC STUDIES & LABORATORY TRIAGE

Urinalysis, microscopy, urine culture, BMP, coagulation panel. CT Urography (gold standard for non-glomerular upper tract) or Renal US. Cystoscopy for lower tract assessment.

MANAGEMENT & EMERGENCY RESUSCITATION PROTOCOLS

Address clot retention: large-bore 3-way Foley catheter (22-24 Fr) with manual irrigation using saline until clear, followed by Continuous Bladder Irrigation (CBI). Treat infection or stones. Urologic outpatient referral for cystoscopy in all unexplained gross hematuria.



HIGH-YIELD BOARD PEARL: Painless gross hematuria in any patient over 40 years old is bladder cancer until proven otherwise and mandates urgent outpatient urologic evaluation with cystoscopy and CT urography.

Topic 8: Hernias (Inguinal, Femoral & Incarceration)

EMERGENCY MEDICINE CLINICAL PRACTICE & BOARD SPECIFICATION

PATHOPHYSIOLOGY & ETIOLOGY

Protrusion of viscus through a defect. Inguinal: Indirect (lateral to inferior epigastric vessels through internal inguinal ring into scrotum; patent processus vaginalis) vs Direct (medial to epigastrics in Hesselbach triangle). Femoral: below inguinal ligament (highest risk of strangulation).

CLINICAL PRESENTATION & PHYSICAL EXAM

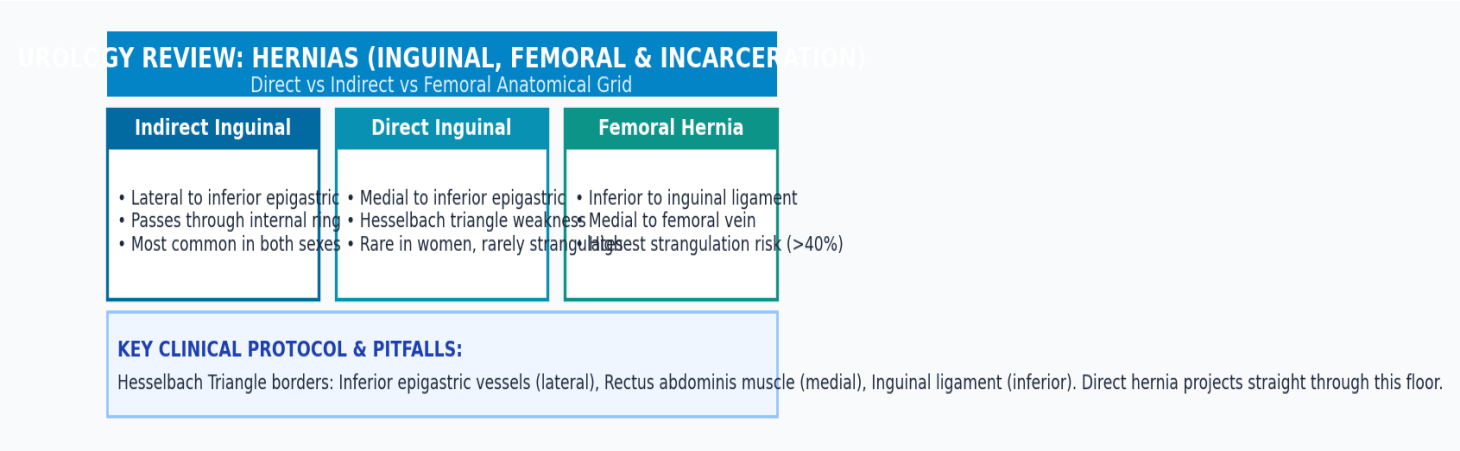
• **Presentation:** Bulge in groin or scrotum. Reducible: painless, disappears when supine. Incarcerated: non-reducible, painful. Strangulated: severe ischemic pain, erythema over bulge, systemic toxicity, bowel obstruction.
• **Physical Exam:** Palpable groin bulge. Test for reducibility. Strangulation signs: skin erythema, warmth, exquisite tenderness, signs of peritonitis.

DIAGNOSTIC STUDIES & LABORATORY TRIAGE

Clinical diagnosis. Bedside ultrasound shows peristalsis or lack of blood flow in strangulated bowel. Non-contrast or contrast CT Abdomen/Pelvis confirms incarceration/strangulation.

MANAGEMENT & EMERGENCY RESUSCITATION PROTOCOLS

Reducible: outpatient elective surgical repair. Incarcerated without toxicity: gentle manual reduction (ice, sedation, Trendelenburg). Strangulated (erythema, fever, peritonitis): NEVER attempt reduction; NPO, IV crystalloids, broad-spectrum IV antibiotics, emergent surgical exploration.



HIGH-YIELD BOARD PEARL: Femoral hernias occur below the inguinal ligament and have the highest rate of incarceration and strangulation (>40%); they are more common in elderly females. Never forcefully reduce a strangulated hernia due to risk of reducing gangrenous bowel into the peritoneal cavity.

Topic 9: Urinary Incontinence (Emergency Evaluation)

EMERGENCY MEDICINE CLINICAL PRACTICE & BOARD SPECIFICATION

PATHOPHYSIOLOGY & ETIOLOGY

Involuntary leakage of urine. Urge (detrusor overactivity), Stress (pelvic floor weakness, urethral hypermobility), Overflow (impaired detrusor contractility or outlet obstruction: BPH, fecal impaction, neurogenic).

CLINICAL PRESENTATION & PHYSICAL EXAM

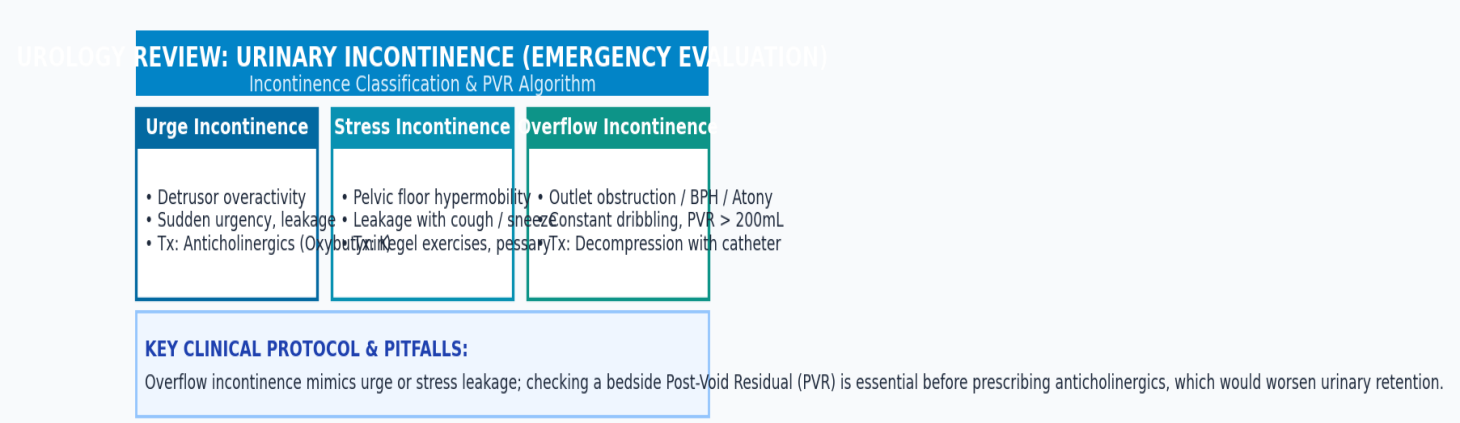
• **Presentation:** Urge: sudden intense desire to void followed by involuntary leakage. Stress: leakage with coughing, sneezing, laughing. Overflow: constant dribbling, fullness, high post-void residual.
• **Physical Exam:** Abdominal exam for distended bladder. Neurological exam (perianal sensation, anal sphincter tone, lower extremity reflexes) to rule out Cauda Equina Syndrome. Pelvic / prostate exam.

DIAGNOSTIC STUDIES & LABORATORY TRIAGE

Urinalysis to rule out UTI. Post-Void Residual (PVR) via bedside bladder scan or in-and-out catheterization (PVR > 200 mL indicates overflow incontinence / retention).

MANAGEMENT & EMERGENCY RESUSCITATION PROTOCOLS

Overflow / Retention: Decompress with Foley catheter. Urge: Bladder training, pelvic floor exercises, Anticholinergics (Oxybutynin) or Beta-3 agonists (Mirabegron). Stress: Kegel exercises, pessaries. Neurologic acute retention: STAT MRI spine for Cauda Equina.



HIGH-YIELD BOARD PEARL: In any patient presenting with new-onset urinary incontinence or retention with overflow, always test perianal sensation (saddle anesthesia) and rectal sphincter tone to exclude Cauda Equina Syndrome.

Topic 10: Nephrolithiasis & Urolithiasis

EMERGENCY MEDICINE CLINICAL PRACTICE & BOARD SPECIFICATION

PATHOPHYSIOLOGY & ETIOLOGY

Crystal supersaturation in urinary tract. Types: Calcium oxalate (80%, radiopaque), Calcium phosphate (radiopaque), Struvite (magnesium ammonium phosphate, staghorn calculi, Proteus mirabilis), Uric acid (radiolucent on plain X-ray), Cystine.

CLINICAL PRESENTATION & PHYSICAL EXAM

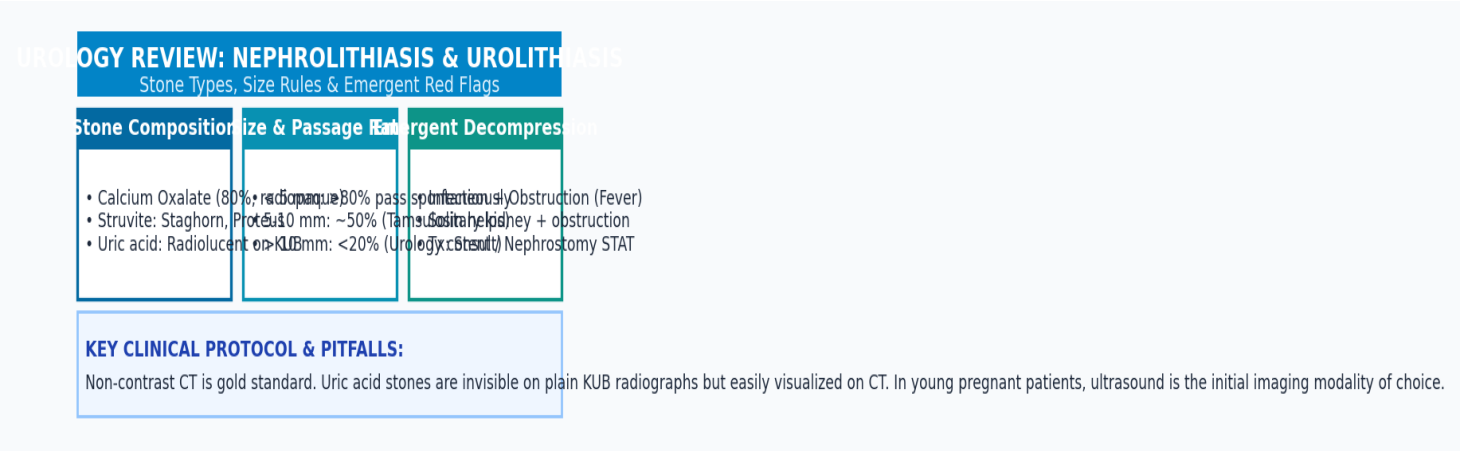
• **Presentation:** Acute, severe, colicky flank pain radiating to groin, labia, or testicle. Nausea, vomiting, hematuria, writhing in pain (cannot sit still).
• **Physical Exam:** CVA tenderness, writhing agitation, unremarkable abdominal exam without peritonitis.

DIAGNOSTIC STUDIES & LABORATORY TRIAGE

Non-contrast helical CT Abdomen/Pelvis (gold standard; identifies size, location, hydronephrosis). Bedside Renal Ultrasound (POCUS) detects hydronephrosis and avoids radiation. Urinalysis: microscopic hematuria (absent in 10-15%).

MANAGEMENT & EMERGENCY RESUSCITATION PROTOCOLS

Analgesia: Ketorolac (IV NSAID, first line) + Opioids for breakthrough. Hydration, Antiemetics. Medical Expulsive Therapy (Tamsulosin 0.4 mg daily for stones 5-10 mm). Emergency Urology consult: Obstructed infected kidney (stone + fever/pyuria = nephrostomy or stent), solitary kidney with obstruction, refractory pain.



HIGH-YIELD BOARD PEARL: Urolithiasis with concurrent urinary tract infection (stone + fever, pyuria, or bacteremia) is a surgical emergency requiring emergent decompression via retrograde ureteral stenting or percutaneous nephrostomy; antibiotics alone are insufficient.

Topic 11: Acute Orchitis

EMERGENCY MEDICINE CLINICAL PRACTICE & BOARD SPECIFICATION

PATHOPHYSIOLOGY & ETIOLOGY

Inflammation of one or both testes. Most viral cause: Mumps virus (Paramyxovirus, occurs in 20-30% of postpubertal males with mumps). Bacterial causes: extension from epididymitis (C. trachomatis, N. gonorrhoeae, E. coli).

CLINICAL PRESENTATION & PHYSICAL EXAM

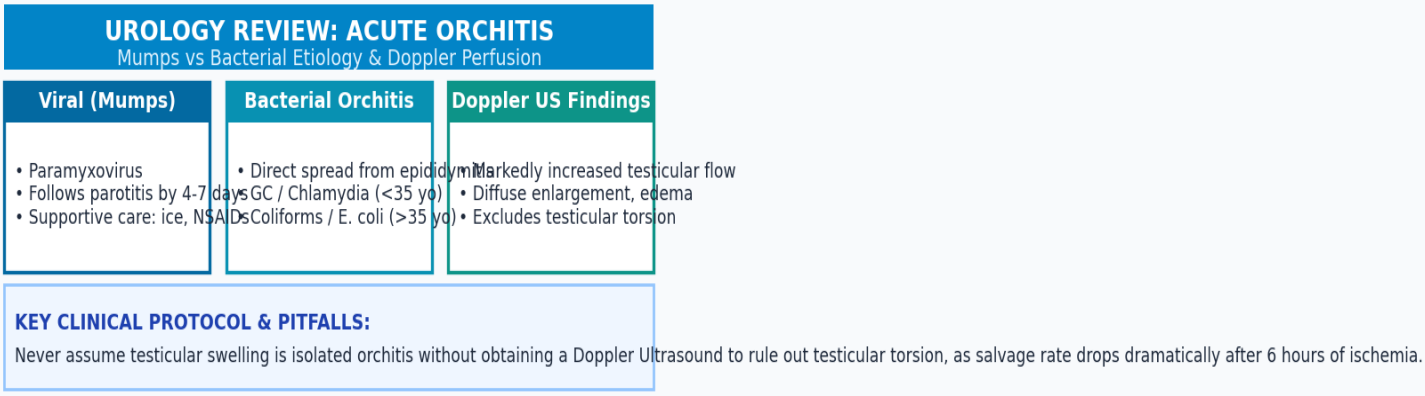
• **Presentation:** Testicular pain, swelling, erythema, fever, chills, parotid swelling 4-7 days prior in mumps orchitis.
• **Physical Exam:** Testicular tenderness, enlargement, induration, scrotal wall erythema. Cremasteric reflex is intact.

DIAGNOSTIC STUDIES & LABORATORY TRIAGE

Scrotal color Doppler US (demonstrates testicular hyperemia and normal or increased blood flow; essential to exclude testicular torsion). Urinalysis, Mumps IgM.

MANAGEMENT & EMERGENCY RESUSCITATION PROTOCOLS

Mumps orchitis: supportive care (bed rest, scrotal support, ice, NSAIDs/analgesia). Bacterial orchitis: treated as epididymo-orchitis (Ceftriaxone + Doxycycline for young males; Levofloxacin for older males).



HIGH-YIELD BOARD PEARL: Mumps orchitis typically occurs 4-7 days after parotitis. While testicular atrophy can develop in up to 50% of affected testes, absolute sterility is rare, though subfertility is common with bilateral involvement.

Topic 12: Acute Bacterial Prostatitis

EMERGENCY MEDICINE CLINICAL PRACTICE & BOARD SPECIFICATION

PATHOPHYSIOLOGY & ETIOLOGY

Acute infection of the prostate gland. Pathogens: E. coli (80%), Klebsiella, Proteus, Pseudomonas; N. gonorrhoeae & C. trachomatis in sexually active young males.

CLINICAL PRESENTATION & PHYSICAL EXAM

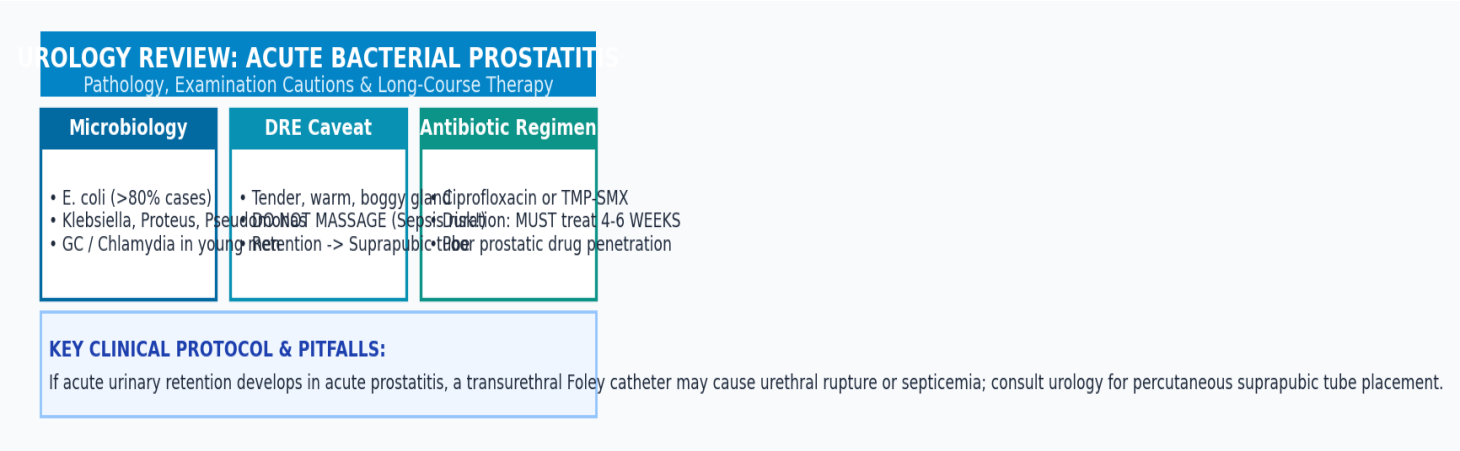
• **Presentation:** Fever, chills, low back and perineal pain, dysuria, urgency, frequency, obstructive urinary symptoms (hesitancy, weak stream, acute urinary retention).
• **Physical Exam:** Exquisitely tender, warm, boggy, swollen prostate on Digital Rectal Exam (DRE). Gentle examination only!

DIAGNOSTIC STUDIES & LABORATORY TRIAGE

Urinalysis with culture (pyuria, bacteriuria). CBC (leukocytosis). PSA is elevated (do not order acutely). Blood cultures if toxic.

MANAGEMENT & EMERGENCY RESUSCITATION PROTOCOLS

Vigorous prostatic massage is STRICTLY CONTRAINDICATED (risks bacteremia and septic shock). Outpatient stable: Ciprofloxacin 500 mg BID OR TMP-SMX DS BID for 4-6 WEEKS. Inpatient toxic: IV Ampicillin + Gentamicin OR Ceftriaxone. Acute retention: suprapubic catheterization (avoid urethral Foley).



HIGH-YIELD BOARD PEARL: Prostatic massage is strictly contraindicated in acute bacterial prostatitis due to the risk of inducing bacteremia and septic shock. Prolonged antibiotic therapy for 4-6 weeks is required due to poor penetration into prostatic tissue.

Topic 13: Acute Pyelonephritis

EMERGENCY MEDICINE CLINICAL PRACTICE & BOARD SPECIFICATION

PATHOPHYSIOLOGY & ETIOLOGY

Ascending infection of the renal parenchyma and pelvicalyceal system. Pathogens: E. coli (75-95%), Klebsiella, Proteus, Enterobacter, Pseudomonas.

CLINICAL PRESENTATION & PHYSICAL EXAM

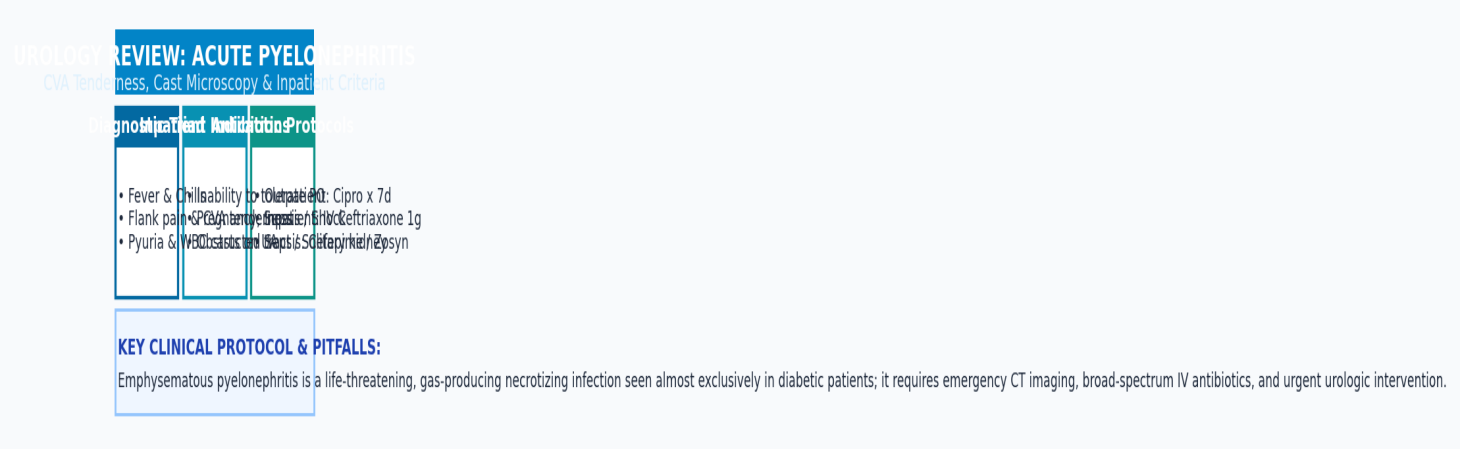
- **Presentation:** Fever (>38°C), shaking chills, flank pain, nausea, vomiting, costovertebral angle tenderness, dysuria, frequency.
- **Physical Exam:** Unilateral or bilateral Costovertebral Angle (CVA) tenderness, tachycardia, fever, warm flushed skin or signs of sepsis.

DIAGNOSTIC STUDIES & LABORATORY TRIAGE

Urinalysis: pyuria, positive nitrites, and WBC casts (pathognomonic for upper tract infection). Urine culture, CBC, BMP. Renal CT with contrast only if complicated, refractory (>48-72h of abx), or suspect abscess/obstruction.

MANAGEMENT & EMERGENCY RESUSCITATION PROTOCOLS

Outpatient (mild/stable): Ciprofloxacin 500 mg BID x 7d OR Levofloxacin 750 mg daily x 5d OR Ceftriaxone 1g IV dose then oral. Inpatient (toxic, pregnant, vomiting, elderly): IV Ceftriaxone 1g daily OR Cefepime OR Piperacillin-tazobactam. Fluid resuscitation.



HIGH-YIELD BOARD PEARL: WBC casts in urine microscopy are pathognomonic for pyelonephritis and distinguish upper UTI from lower UTI (cystitis). Treatment failure after 48-72 hours of appropriate antibiotics warrants CT imaging to rule out perinephric abscess or obstructing calculus.

Topic 14: Testicular Torsion (Urologic Emergency)

EMERGENCY MEDICINE CLINICAL PRACTICE & BOARD SPECIFICATION

PATHOPHYSIOLOGY & ETIOLOGY

Twisting of the spermatic cord resulting in venous occlusion, arterial compromise, and testicular ischemia/necrosis. Predisposing anatomy: "Bell-clapper" deformity (failure of normal posterior fixation of testis and epididymis).

CLINICAL PRESENTATION & PHYSICAL EXAM

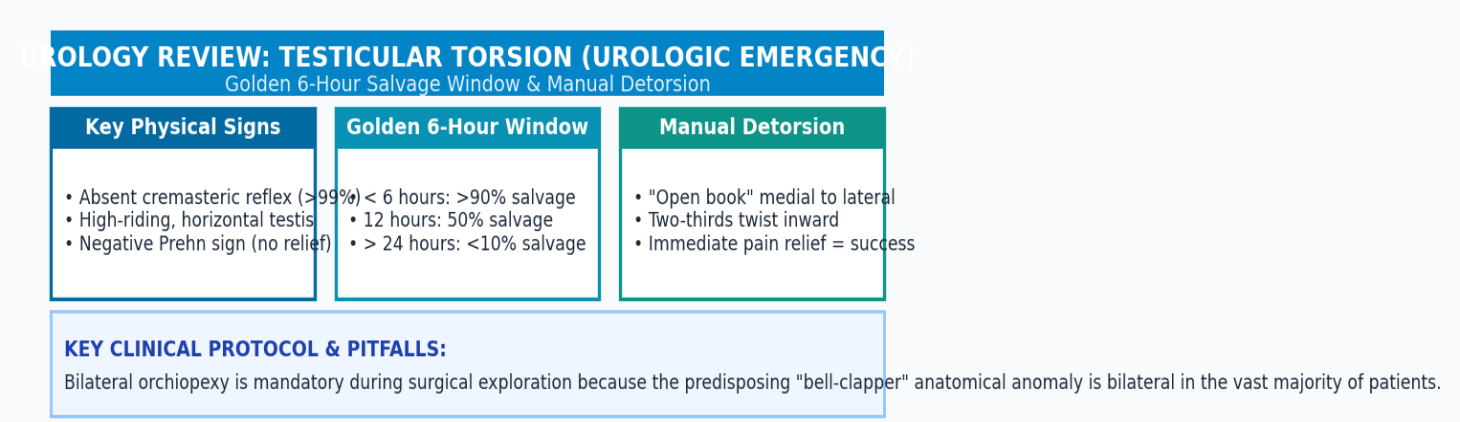
• **Presentation:** Sudden, severe unilateral testicular pain, often awakening patient from sleep, radiating to lower abdomen/groin, accompanied by nausea and vomiting.
• **Physical Exam:** High-riding, horizontally-oriented, swollen, tender testis. Exquisite tenderness. Absence of the cremasteric reflex (most sensitive physical exam finding). Negative Prehn sign.

DIAGNOSTIC STUDIES & LABORATORY TRIAGE

STAT Scrotal Color Doppler Ultrasound (shows absent or decreased intratesticular blood flow and coiled spermatic cord "whirlpool sign"). However, IMAGING MUST NOT DELAY EMERGENCY SURGICAL CONSULTATION.

MANAGEMENT & EMERGENCY RESUSCITATION PROTOCOLS

Emergent Urologic Consultation for Surgical Detorsion and bilateral orchiopexy (salvage rate >90% if within 6 hours; <10% after 24 hours). Manual detorsion: "Open book" maneuver (rotate testis medial to lateral, 180-540 degrees; like opening a book facing patient).



HIGH-YIELD BOARD PEARL: Absence of the cremasteric reflex is the most sensitive physical finding in testicular torsion (>99%). Time is testicular tissue: salvage rate is >90% if detorsion occurs within 6 hours of symptom onset. Never delay surgical consultation for ultrasound confirmation.

Topic 15: Acute Urethritis (Gonococcal vs Non-Gonococcal)

EMERGENCY MEDICINE CLINICAL PRACTICE & BOARD SPECIFICATION

PATHOPHYSIOLOGY & ETIOLOGY

Infection and inflammation of the urethra. Gonococcal Urethritis (GU): Neisseria gonorrhoeae (Gram-negative intracellular diplococci). Non-Gonococcal Urethritis (NGU): Chlamydia trachomatis (most common), Mycoplasma genitalium, Trichomonas vaginalis.

CLINICAL PRESENTATION & PHYSICAL EXAM

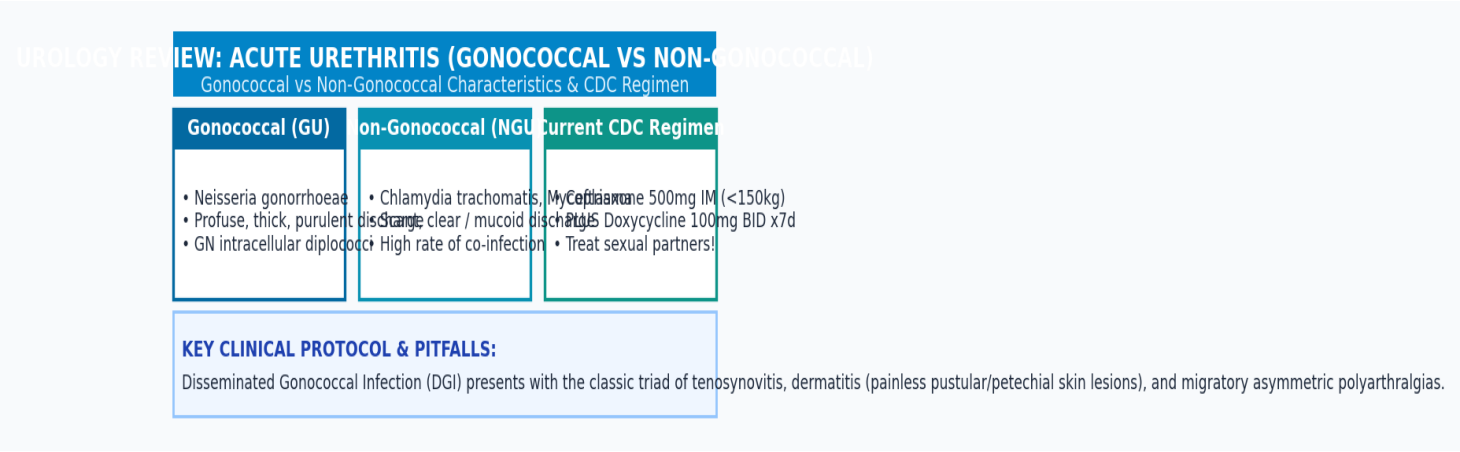
• **Presentation:** Urethral discharge (profuse purulent in gonorrhea; scant, watery/mucoid in Chlamydia), dysuria, urethral pruritus.
• **Physical Exam:** Penile exam shows urethral erythema and active discharge. Palpate for meatal stenosis or testicular tenderness.

DIAGNOSTIC STUDIES & LABORATORY TRIAGE

Urethral swab or first-catch dirty urine NAAT (Nucleic Acid Amplification Test) for N. gonorrhoeae and C. trachomatis. Gram stain of discharge: Gram-negative intracellular diplococci.

MANAGEMENT & EMERGENCY RESUSCITATION PROTOCOLS

Empiric Dual Therapy: Ceftriaxone 500 mg IM single dose (for gonorrhea, 1g if >= 150 kg) PLUS Doxycycline 100 mg PO BID x 7 days (for Chlamydia). Treat all sexual partners. Patient must abstain from sexual activity for 7 days post-treatment.



HIGH-YIELD BOARD PEARL: Gram stain demonstrating Gram-negative intracellular diplococci inside polymorphonuclear leukocytes confirms Neisseria gonorrhoeae. Always co-treat for Chlamydia with Doxycycline 100 mg BID x 7 days unless Chlamydia is excluded by NAAT.